

Detecting Multilingual, Multicultural And Multievent Online Polarization

Dodun-des-Perrieres Dan, Gheorghita Stefana, Ivanusca Teodor

Abstract

We present our system for the SemEval-2026 POLAR shared task on *binary polarization detection* across 22 languages and multiple social media platforms. The dataset exhibits strong cross-lingual heterogeneity: while globally near-balanced, individual languages range from heavily polarized to overwhelmingly non-polarized, motivating the use of **macro-F1** as the primary metric.

Our approach follows a staged strategy. We first study English in a controlled monolingual setting to establish baselines and validate modeling choices, then extend to multilingual training. In English, bag-of-words baselines (Naive Bayes and Logistic Regression) provide a lower bound and reveal a bias toward the majority class. Fine-tuning contextual encoders substantially improves performance: RoBERTa-base already reaches **0.79** accuracy and **0.76** macro-F1, and scaling to RoBERTa-large combined with GPT-based paraphrase augmentation further improves validation performance to **0.83** accuracy and **0.81** macro-F1.

In the multilingual setting, we evaluate XLM-RoBERTa, MPNet sentence encoders, and mDeBERTa-v3 variants under different optimization and imbalance-aware strategies, observing that gains are highly language-dependent and that no single configuration dominates across all languages. Overall, our findings highlight the importance of macro-averaged evaluation under cross-lingual imbalance and show that transformer fine-tuning with targeted augmentation can yield consistent improvements for polarization detection.

1 Introduction

The rapid expansion of social media platforms has had a substantial impact on interpersonal communication. While these platforms facilitate information exchange, they also contribute to the emergence and reinforcement of polarized narratives, which

can intensify social divisions and hinder constructive dialogue. As a result, identifying and analyzing polarization in online messages has become an important challenge for natural language processing (NLP) research.

Recent advances in transformer-based language models have led to substantial improvements across a wide range of NLP tasks, including text classification, sentiment analysis, hate speech, and stance detection. These models provide powerful contextual representations that are well suited for capturing subtle linguistic cues associated with polarization. However, polarization is inherently influenced by linguistic, cultural, and socio-political factors, which vary considerably across languages and regions. This makes multilingual and multicultural polarization detection particularly challenging.

To support research in this area, (Naseem et al., 2026) introduces *POLAR* a multilingual dataset designed to capture diverse forms of polarized content across multiple languages and cultural contexts. The task encourages the development of robust models capable of generalizing beyond language-specific patterns and addressing real-world multilingual scenarios. In this work, we investigate both monolingual and multilingual approaches to polarization detection, analyzing their strengths and limitations under varying data conditions.

2 State-of-the-Art

Early work on polarization and related phenomena in online discourse has focused on tasks such as hate speech, stance, and offensive language detection, often within monolingual and event-specific settings. These studies typically relied on supervised learning methods and targeted explicit signals of hostility or strong opinion polarity in social media texts.

More recently, transformer-based language models have become the dominant approach for treating

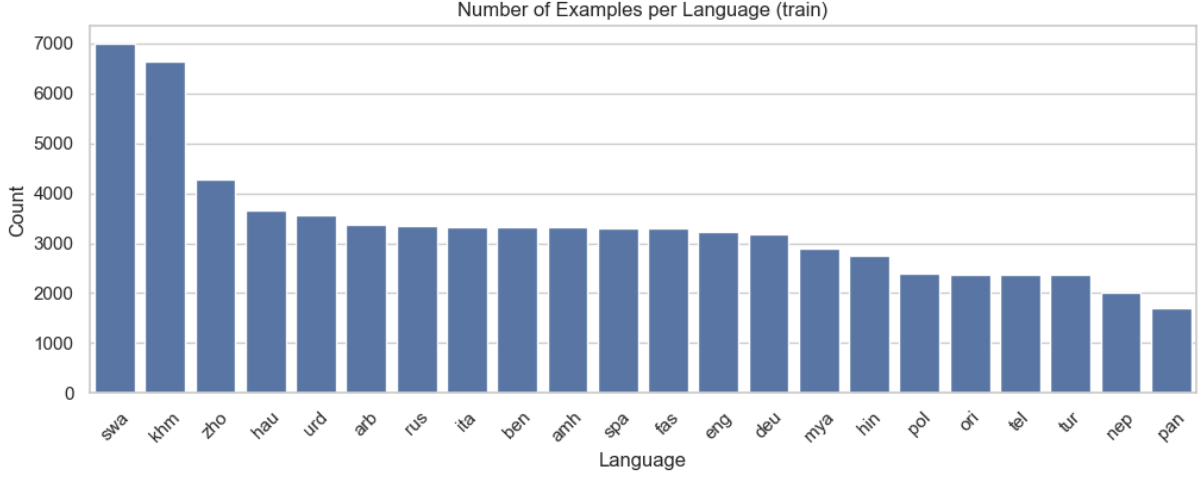


Figure 1: Train Samples per Language

these tasks. Multilingual pretrained models such as mBERT, XLM-R, and mDeBERTa have demonstrated strong performance in cross-lingual text classification tasks (Conneau et al., 2020), (Arango et al., 2021), enabling transfer learning across languages with limited annotated data.

The POLAR benchmark (Naseem et al., 2026) dataset covers 22 languages and annotates texts along three complementary dimensions: binary polarization detection, polarization type, and polarization manifestation. Experimental results reported by the dataset authors show that multilingual transformer models achieve reasonable performance on binary detection but exhibit substantial performance degradation on fine-grained tasks such as type and manifestation classification, especially in cross-lingual evaluation scenarios. These findings highlight the inherent complexity of polarization as a socially and culturally grounded phenomenon.

Related research has also explored multi-task learning as a means of improving performance on closely related objectives, such as jointly modeling hate speech, stance, and target detection (Rodriguez-Garcia and Centeno, 2024), (Kang et al., 2025). Multi-task transformer architectures have been shown to benefit from shared representations across tasks, particularly when annotated data is limited. Additionally, domain-specific multilingual representations have been proposed to better capture culturally sensitive language patterns, outperforming general-purpose multilingual embeddings in zero-shot transfer settings (Arango et al., 2021).

Overall, existing work suggests that while large multilingual models provide a strong foundation for

polarization detection, effectively modeling the linguistic diversity, cultural specificity, and rhetorical strategies involved in polarized discourse remains an open challenge.

3 Problem Description

This work addresses the SemEval 2026 POLAR shared task (Naseem et al., 2026), which aims to automatically identify polarized content in online messages across multiple languages and cultural contexts. Given a short text extracted from social media platforms, systems are required to determine whether the text contains polarization and, depending on the subtask, further characterize its nature.

3.1 Task Definition

The POLAR dataset defines three complementary subtasks: binary polarization detection, polarization type classification, and polarization manifestation identification.

In this work, we focus on the *binary polarization detection* subtask. For each input message, the system predicts whether the content is *Polarized* or *Not Polarized*.

3.2 Dataset

The dataset includes 22 languages spanning diverse linguistic and regional contexts, with data sourced from multiple platforms such as Twitter/X, Facebook, BlueSky, Reddit, and local news outlets (Naseem et al., 2026). Each instance consists of a unique identifier, the input text, and a binary polarization label.

The languages are represented in different proportions across the dataset, as it can be seen in

Figure 1.

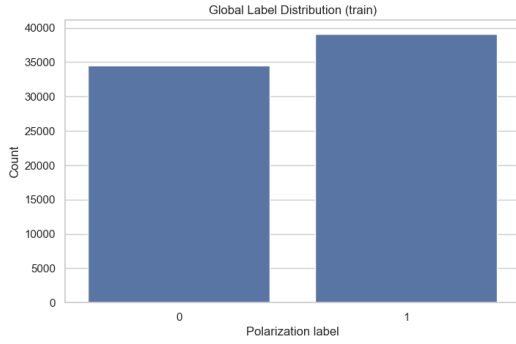


Figure 2: Label Distribution

Although the full dataset, comprising 73,681 entries, is approximately balanced with respect to polarization labels (Figure 2), the label distribution varies substantially between languages, which strongly affects model performance and results, as illustrated in Figure 3.

3.3 Exploratory Data Analysis

To better characterize the challenges of multilingual polarization detection, we conduct an exploratory analysis of the training data, focusing on cross-lingual label distributions, text length properties, and their relationship with the polarization label.

Cross-lingual label imbalance. While the dataset is approximately balanced at the global level, the proportion of polarized posts varies substantially across languages. Figure 4 illustrates this variability, with polarization rates ranging from around 10% in Hausa to over 90% in Khmer. Several languages exhibit near-balanced distributions (e.g., English, Spanish, Swahili), whereas others are strongly skewed toward one class. This pronounced heterogeneity suggests that polarization manifests differently across linguistic and cultural contexts and motivates the use of macro-averaged evaluation metrics to avoid bias toward languages with extreme class imbalance.

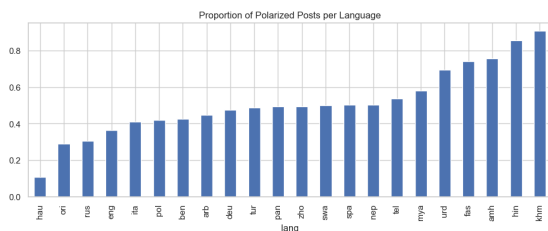


Figure 4: Proportion of polarized posts per language.

Text length by polarization label. We analyze text length in terms of word counts across the english data. Figure 5 shows that polarized posts tend to be slightly longer on average than non-polarized ones; however, the overlap between the two distributions is substantial. Although a small number of long outliers are present in both classes, most texts remain relatively short. This indicates that while length may weakly correlate with polarization, it is insufficient as a standalone predictive signal.

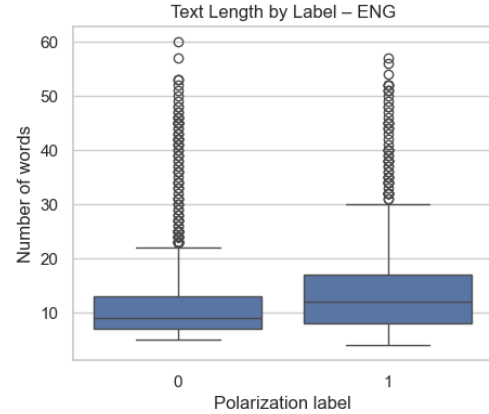


Figure 5: Text length (in words) by polarization label.

Length-feature correlations. To further assess the informativeness of surface-level features, we compute correlations between polarization, character length, and word length. As shown in Figure 6, character and word counts are strongly correlated with each other, while their correlation with the polarization label is weak (absolute correlation below 0.1). This confirms that surface-level length features alone do not capture the complexity of polarization and highlights the need for models that leverage semantic and contextual information.

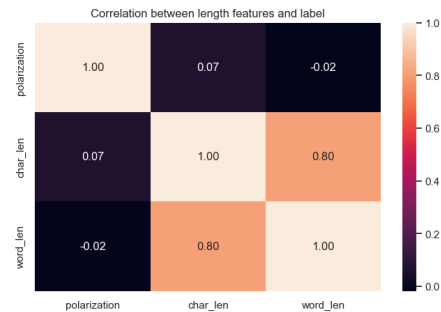


Figure 6: Correlation between polarization and length-based features.

Cross-lingual variation in text length. Text length distributions differ markedly across lan-

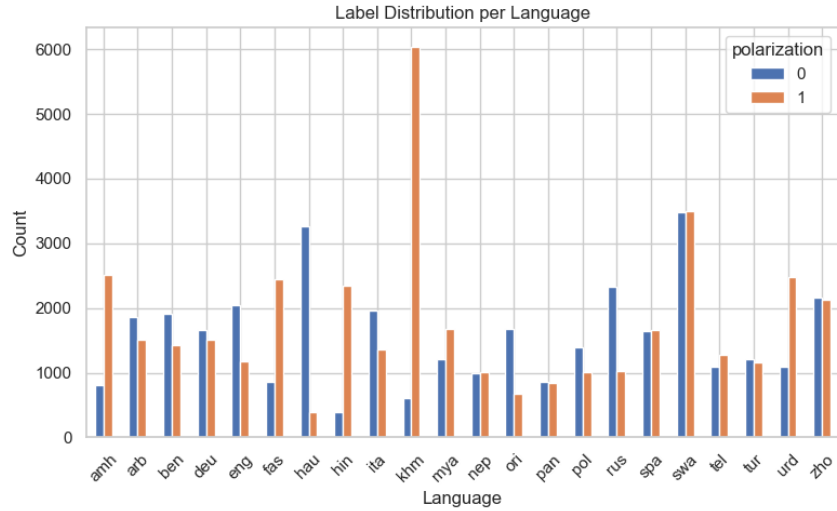


Figure 3: Label Distribution per Language

guages, as illustrated in Figure 7. Languages such as Bengali, Hindi, and Russian exhibit longer median word counts, while others, including Chinese and Khmer, contain substantially shorter texts due to differences in writing systems and tokenization granularity. These discrepancies underline the linguistic diversity of the dataset and pose an additional challenge for multilingual modeling, as a single representation must accommodate highly heterogeneous input characteristics.

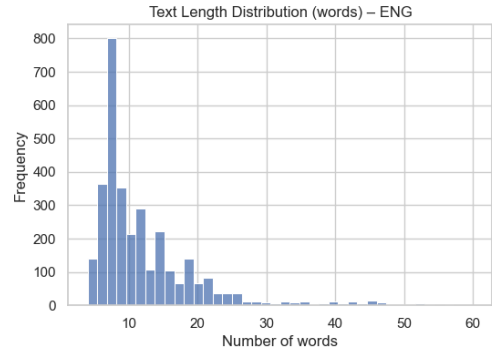


Figure 8: Word-length distribution for English.

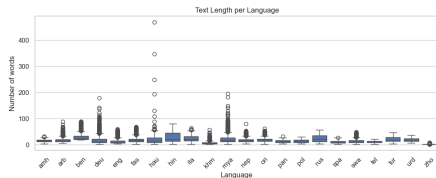


Figure 7: Word Len by language

Language-specific length distributions. Figure 8 presents the word-length distribution for English as an illustrative example. The distribution is heavily right-skewed, with most texts containing fewer than 20 words and a long tail of rare but very long posts. Similar patterns are observed across languages, reflecting the social-media-driven nature of the data. The prevalence of short texts increases ambiguity, while the presence of long outliers motivates the use of input truncation during training to ensure computational efficiency and robustness.

Overall, this exploratory analysis highlights three key aspects of the dataset: strong cross-lingual variation in label distributions, substantial heterogeneity in text length across languages, and the limited predictive power of surface-level length features. These observations motivate the adoption of multilingual, contextual models capable of capturing semantic cues beyond simple lexical or structural statistics.

4 Methodology

4.1 Overview and Experimental Strategy

We adopt a staged experimental strategy to address the polarization detection task. Our approach begins in a monolingual setting, focusing exclusively on English, before extending to a multilingual scenario. This design choice is motivated by the relative simplicity and interpretability of the monolingual setting, which allows us to validate modeling assumptions, establish meaningful baselines, and analyze model behavior in a controlled environ-

ment.

Starting with English enables us to determine whether polarization can be effectively captured using simple lexical cues or whether deeper semantic representations are required. Once robust modeling choices are identified in this setting, we scale the approach to multiple languages, where additional challenges such as cross-lingual variation, heterogeneous label distributions, and linguistic diversity arise. This progressive strategy ensures that complexity is introduced incrementally and that decisions made in the multilingual setting are grounded in validated monolingual experiments.

4.2 Data Preprocessing

The dataset provided by the task organizers requires minimal preprocessing. The following steps are applied consistently across all experiments:

- **Data integrity.** All instances contain valid text and label annotations. No missing values are present in the dataset.
- **Text normalization.** No additional normalization or filtering is applied. Explicit social media artifacts such as URLs, mentions, hashtags, and emojis occur infrequently and are retained in their original form.
- **Tokenization.** Texts are tokenized using the tokenizer associated with each respective model.
- **Sequence length.** Input sequences are truncated to a fixed maximum length during training to ensure computational efficiency and consistent batch shapes.

4.3 Data Augmentation.

We experiment with data augmentation techniques to increase the diversity of minority-class instances. The following approaches are evaluated:

- **Back-Translation.** Paraphrases are generated using an English–Spanish–English pipeline based on Helsinki-NLP translation models. The resulting back-translated texts are used to oversample the minority class.
- **LLM-Based Paraphrasing.** We employ the OpenAI GPT-4.1 API to generate paraphrases of the original texts while preserving both semantic content and polarization label. Generated samples are added to the training data to increase minority-class representation.

- **LLM-Based Generation.** We employ the gemini-2.5-flash-lite API to generate new entries to balance the datasets using few-shot prompting. Generated samples are added to the training data to increase minority-class representation.

Augmented samples are used exclusively during training and are never included in validation or evaluation splits.

4.4 Monolingual Experiments (English)

We first address polarization detection in a monolingual setting by focusing exclusively on English. This setting provides a controlled environment for validating modeling assumptions and analyzing the effectiveness of different approaches before extending to multilingual models.

4.4.1 Baselines

As an initial point of reference, we implement classical machine learning baselines using Naive Bayes and Logistic Regression. Both models operate on bag-of-words representations and serve as lightweight baselines to assess whether polarization can be captured using shallow lexical cues alone. While these approaches provide a useful sanity check, their limited capacity motivates the use of contextualized representations.

4.4.2 Monolingual Modeling Progression

Following the baseline experiments, we focused our investigation on contextual encoder models, with particular emphasis on RoBERTa. Initial experiments were conducted using RoBERTa-base in a single train–validation split setting to assess the suitability of contextual representations for polarization detection and to rapidly iterate on modeling choices. Based on consistent performance improvements, we subsequently adopted RoBERTa-large, which offers greater representational capacity for modeling nuanced semantic cues present in polarized content.

For comparison, we also evaluated DeBERTa-v3 under the same experimental conditions. While DeBERTa achieved competitive performance, it did not consistently outperform RoBERTa-large in the monolingual setting. As a result, RoBERTa-large was selected as the primary architecture for subsequent experiments.

In addition to encoder fine-tuning, we briefly explored large language model–based classification

using the GPT-4.1 API. Despite careful prompt design, this approach produced unstable predictions and inferior performance compared to fine-tuned encoder models, and was therefore not pursued further.

4.4.3 Strongest Monolingual Model

Based on validation results, RoBERTa-large emerges as the strongest monolingual model and is selected as the final base architecture for English polarization detection. All subsequent experiments in the monolingual setting build upon this model.

Training Setup. Early experiments relied on a single train-validation split for efficiency during model selection. Once RoBERTa-large was identified as the most promising architecture, we transitioned to a 5-fold stratified cross-validation strategy using StratifiedKFold to improve robustness and reduce variance in performance estimates. Models trained on each fold are used to generate predictions, which are then aggregated using a soft-voting ensemble by averaging the predicted class probabilities. Training is performed using the AdamW optimizer and cross-entropy loss.

Handling Class Imbalance. Although the English subset is less skewed than several other languages, it still exhibits a noticeable class imbalance (2,047 non-polarized vs. 1,175 polarized instances). To mitigate bias toward the majority class, we incorporate class weights into the loss function, increasing the penalty for misclassification of minority-class instances. This approach encourages more balanced decision boundaries without altering the underlying data distribution. We further evaluate data augmentation strategies to assess whether increasing minority-class diversity yields additional gains.

4.4.4 Additional Experiments

In addition to the primary RoBERTa-based classifier, we conducted several exploratory experiments to assess alternative modeling and optimization strategies.

Sentence-Level Representations. We evaluated a sentence-embedding-based classifier using sentence-transformers/all-mpnet-base-v2, where token representations are aggregated via mean pooling before classification (Figure 9). This approach simplifies the representation to a single sentence vector, but consistently underperformed

token-level fine-tuning, indicating that polarization cues benefit from richer token-level interactions.

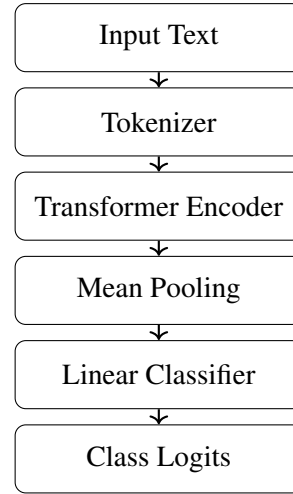


Figure 9: Sentence-Embedding-Based Classification Architecture with Mean Pooling.

Large Language Model Classification. We also experimented with prompt-based classification using the GPT-4.1 API. Despite careful prompt engineering and multiple prompting strategies, this approach resulted in unstable predictions and inferior performance compared to fine-tuned encoder models. Given its sensitivity to prompt formulation and lack of consistency across runs, this direction was not pursued further.

Optimization Variants. Finally, we explored layer-wise learning rate decay (LLRD) as an optimization strategy for fine-tuning RoBERTa. This approach assigns higher learning rates to upper transformer layers and progressively smaller rates to lower layers, aiming to preserve general linguistic knowledge while adapting higher-level representations. Although LLRD led to faster convergence in some settings, it did not consistently improve validation performance over a uniform learning rate and was therefore not adopted in the final configuration.

4.4.5 Hybrid Monolingual Model

Interestingly, although sentence-level embeddings alone did not perform competitively on the English dataset, we observed a notably stronger performance for the MPNet-based sentence encoder in the multilingual setting, where it achieved a macro-F1 score of 81.2 for English. This observation suggested that sentence-level semantic representations may be particularly beneficial in multilingual scenarios, where cross-lingual alignment and

global semantic cues play a more prominent role. Motivated by this result, and by the broader observation that different transformer models capture complementary linguistic information, we explored a hybrid architecture that integrates sentence-level representations from MPNet with token-level representations from RoBERTa (Figure 10).

In this hybrid architecture, the input text is processed in parallel by two branches. In the first branch, the text is encoded using RoBERTa, and the final [CLS] representation is extracted as a sequence-level summary, capturing rich token-level contextual information. In the second branch, the same input is passed through an MPNet encoder, and the resulting token embeddings are aggregated via mean pooling to obtain a sentence-level representation. This embedding is subsequently L2-normalized and projected through a linear layer in order to align its dimensionality with the RoBERTa representation. The two representations are then concatenated and fed into a lightweight multilayer perceptron classifier, which produces the final prediction logits.

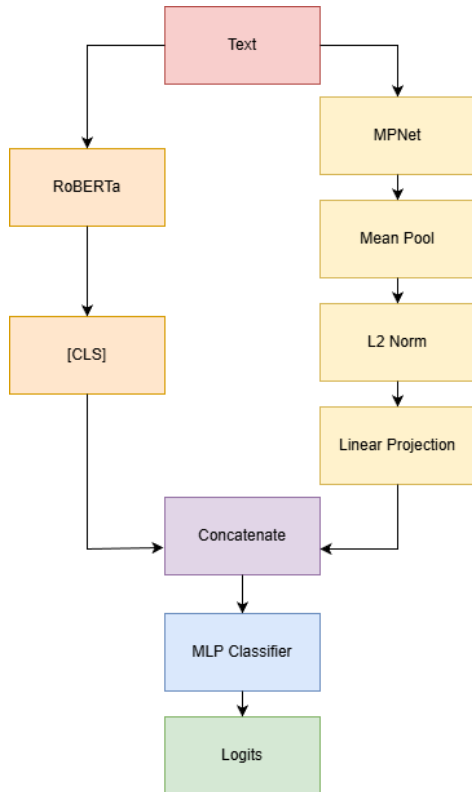


Figure 10: Hybrid Model

4.5 Multilingual Classification

For multilingual classification, we employ multilingual transformer-based models, including XLM-

RoBERTa and mDeBERTa, under different training settings, which will be described in detail in the following sections.

4.5.1 Cost-Sensitive Learning via Weighted Cross-Entropy

To mitigate the bias inherent in training on imbalanced datasets, we employed a cost-sensitive learning approach known as weighted training. Standard empirical risk minimization often allows the classifier to achieve high accuracy by trivially predicting the majority class, thereby neglecting the minority class of interest. By introducing class-specific coefficients into the loss function, we penalize the misclassification of minority examples more heavily than majority ones (King and Zeng, 2001).

For a multi-class problem with C classes, the Weighted Cross-Entropy (WCE) loss for a sample belonging to class c is defined as:

$$L_{\text{WCE}}(y, \hat{y}) = - \sum_{c=1}^C w_c y_c \log(\hat{y}_c) \quad (1)$$

where y_c is the binary indicator (0 or 1) if class label c is the correct classification, $\hat{y}_c$ is the predicted probability, and w_c is the weight assigned to class c , typically inversely proportional to class frequency.

4.5.2 Hard Example Mining with Focal Loss

While weighted cross-entropy addresses class imbalance, it treats all examples within a class equally. To further refine the model’s focus, we integrated Focal Loss (Lin et al., 2018). This technique modifies the standard cross-entropy loss by adding a modulating factor that reduces the loss contribution of “easy” examples (those classified with high confidence). This mechanism prevents the vast number of easy negatives from overwhelming the gradient, directing the model’s learning capacity toward “hard” examples.

The Focal Loss is formally defined as:

$$L_{\text{Focal}}(p_t) = -\alpha_t (1 - p_t)^\gamma \log(p_t) \quad (2)$$

Here, p_t represents the model’s estimated probability for the ground truth class. The focusing parameter $\gamma \geq 0$ controls the rate at which easy examples are down-weighted. As γ increases, the effect of the modulating factor $(1 - p_t)^\gamma$ intensifies, effectively “silencing” well-classified examples and focusing the optimization on the decision boundary.

4.5.3 Semantic Data Augmentation via LLMs

To enhance the generalization capability of the model and address data scarcity in specific sub-domains, we utilized Large Language Models (LLMs) for semantic data augmentation with as well with the gemini-2.5-flash-lite API. Traditional NLP augmentation techniques (e.g., synonym replacement) often degrade the semantic integrity of the text. In contrast, generative augmentation leveraging LLMs allows for the synthesis of high-quality, contextually diverse paraphrases and novel training samples that preserve the original label semantics (Ding et al., 2020; Dai et al., 2023). This approach acts as a regularizer, exposing the model to a wider variance of linguistic structures and reducing overfitting to the specific phrasing of the source dataset.

5 Experimental Setup and Results

5.1 Evaluation Metrics

We evaluate all systems using classification metrics computed from the confusion matrix: true positives (TP), true negatives (TN), false positives (FP), and false negatives (FN), where the positive class corresponds to the polarized label. Since class distributions vary substantially across languages, our primary goal is to measure performance in a way that does not overly favor the majority class.

Per-class Precision, Recall, and F1. For each class $c \in \{0, 1\}$, we compute:

- **Precision** (how many predicted c are correct):

$$P_c = \frac{TP_c}{TP_c + FP_c}$$

- **Recall** (how many true c are recovered):

$$R_c = \frac{TP_c}{TP_c + FN_c}$$

- **F1 score** (harmonic mean of precision and recall):

$$F1_c = \frac{2P_cR_c}{P_c + R_c}$$

We report F1 because it balances precision and recall and is more informative than accuracy under imbalance.

Macro-F1 (Primary Metric). Our main evaluation metric is **macro-averaged F1**, which computes the F1 score independently for each class and then averages them:

$$F1_{\text{macro}} = \frac{1}{2} (F1_0 + F1_1)$$

Macro-F1 assigns equal weight to both classes regardless of their frequency, making it suitable for this task where some languages exhibit strong skew toward one label. In practice, macro-F1 penalizes systems that perform well on the majority class but fail to identify the minority class.

Accuracy (Secondary Metric). For completeness, we also compute accuracy:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

However, accuracy can be misleading in imbalanced settings and is therefore treated as a secondary metric.

Why Macro-F1. We prioritize macro-F1 for the following reasons:

- **Class imbalance:** Several languages are strongly skewed toward one label, so accuracy and micro-averaged metrics may overestimate performance.
- **Fairness across classes:** Macro-F1 gives equal importance to polarized and non-polarized detection.
- **Robust model selection:** It encourages models that maintain balanced precision and recall rather than exploiting majority-class dominance.

5.2 Experimental Setup

5.2.1 Monolingual Settings (English)

Monolingual experiments are conducted on the English subset of the dataset and evaluated using macro-averaged F1.

Initial Training Configuration. Initial experiments are performed using a single train-validation split (90% / 10%) to guide model selection. The training setup is as follows:

- **Optimizer:** AdamW
- **Learning rate:** 5×10^{-6}

- Scheduler: cosine with warmup ratio 0.10
- Weight decay: 0.01
- Training epochs: 3
- Batch size: 4 (train), 8 (evaluation)
- Gradient accumulation: enabled
- Maximum sequence length: 256
- Label smoothing: 0.05

Cross-Validation Setup. After identifying a stable configuration, we adopt stratified k -fold cross-validation with $k = 5$. Each fold is trained independently using the same hyperparameters, and predictions are aggregated via soft-voting by averaging class probabilities. All reported monolingual results are obtained using this cross-validation ensemble.

5.2.2 Multilingual Settings

As an initial experiment, we fine-tuned XLM-RoBERTa on a multilingual dataset obtained by combining instances from all available languages. This experiment followed the default configuration provided in the official baseline, using a learning rate of 2×10^{-5} and a per-device training batch size of 64. Training was conducted for 5 epochs.

Subsequently, we explored a more refined training configuration to improve performance. In this setup, we applied weight decay (0.01), a warmup ratio of 0.10, and label smoothing with a factor of 0.05. Training was again performed for 5 epochs, using a reduced per-device batch size of 4 combined with gradient accumulation over 4 steps. A lower learning rate of 5×10^{-6} was adopted to stabilize optimization.

We further evaluated this refined configuration on a subset of languages, namely English, Italian, Spanish, German, Russian, and Polish.

Building on these configurations, we also experimented with the sentence-embedding-based approach described in the previous section, using the sentence-transformers/paraphrase-multilingual-mpnet-base-v2 model. This model was evaluated under the first two training configurations described above.

Furthermore, we also test the techniques described in the previous section, using mDeBERTa.

Comparative results in terms of macro-F1 on the test set for the three experimental settings are reported in Table 3.

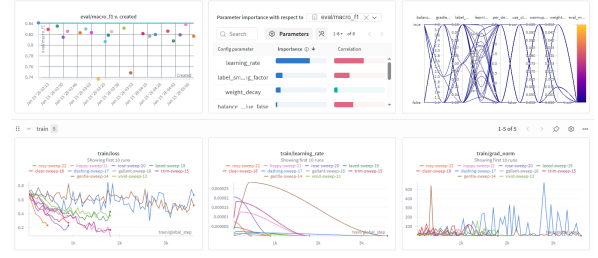


Figure 11: Train Visualizations

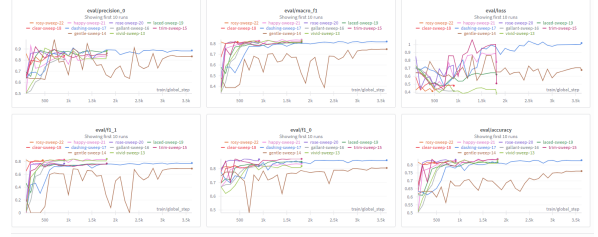


Figure 12: Eval Visualizations

5.3 Experimental Results

5.3.1 Monolingual Results (English)

Baseline Results We first evaluate lightweight baseline models on the English subset to establish a lower bound for polarization detection. As shown in Table 1, Naive Bayes achieves a macro-F1 score of 0.686, while Logistic Regression attains higher accuracy but a lower macro-F1 of 0.652, reflecting a strong bias toward the majority class. Overall, both models struggle to reliably capture polarized content, highlighting the limitations of bag-of-words representations.

Model	Acc.	Macro-F1
Naive Bayes (Count)	0.712	0.686
Logistic Regression (TF-IDF)	0.719	0.652

Table 1: Baseline performance on the English subset.

Hyperparameter Tuning. We performed Bayesian hyperparameter optimization in order to identify effective training configurations for our models. The optimization process was carried



Figure 13: Parameter effect

out using a Bayesian search strategy, with the objective of maximizing the macro-F1 score on the validation set. Each trial corresponded to a full training run, and model performance was evaluated using the validation macro-F1 metric.

The search space covered both optimization-related and regularization-related hyperparameters. The learning rate was sampled from a log-uniform distribution in the range $[10^{-6}, 3 \times 10^{-5}]$, enabling exploration across multiple orders of magnitude. Regularization was controlled via weight decay, with values selected from $\{0.0, 0.01, 0.05\}$, and label smoothing factors from $\{0.0, 0.05, 0.1\}$.

We additionally varied the training dynamics by exploring per-device batch sizes of 4 and 8, combined with gradient accumulation steps of 1, 2, or 4, which allowed effective batch size scaling under GPU memory constraints. Learning rate warmup was applied using warmup ratios of 0.05 and 0.10. Finally, we evaluated strategies for handling class imbalance by toggling the use of class-weighted loss functions and optionally balancing the evaluation split.

This Bayesian optimization setup enabled the search procedure to efficiently focus on promising regions of the hyperparameter space, leading to stable high-performing configurations without requiring an exhaustive grid search.

Several visualizations from the hyperparameter sweep, available at <https://wandb.ai/gheorghitastefana-alexandru-ioan-cuza-university-iasi/DL4NLP/sweeps/jd9rnt8v>, are shown in Figures 11 and 12. In addition, the influence of individual hyperparameters on the validation macro-F1 score is illustrated in Figure 13. Based on the Bayesian hyperparameter optimization, the best-performing configuration was obtained with a learning rate of 7.5×10^{-6} , weight decay set to 0.01, and a label smoothing factor of 0.1. Training was performed using a per-device batch size of 8, a warmup ratio of 0.05, and gradient accumulation set to 1. In addition, both class balancing on the evaluation set and the use of class weights during training were activated. This configuration consistently achieved the highest validation macro-F1 score within the sweep and was therefore selected for subsequent experiments.

Additional Experiments We report intermediate monolingual experiments that guided the development of the final system, focusing on progressively stronger RoBERTa-based configurations.

RoBERTa-base without augmentation. We first fine-tune RoBERTa-base on the English subset without data augmentation. This configuration yields a clear improvement over bag-of-words baselines, achieving an accuracy of **0.79** and a macro-F1 score of **0.76**. Performance remains stronger for the non-polarized class ($F1 = \mathbf{0.84}$) than for the polarized class ($F1 = \mathbf{0.69}$), indicating that contextual representations capture meaningful polarization cues but still struggle with minority-class detection in the absence of explicit imbalance handling.

RoBERTa-large with data augmentation. Building on these results, we adopt RoBERTa-large and augment the training data using GPT-based paraphrasing. This setup further improves performance, reaching an accuracy of **0.83** and a validation macro-F1 score of **0.81**. The observed gains suggest that increased model capacity combined with data augmentation leads to better generalization, particularly for polarized content.

Training dynamics for this configuration are shown in Figures 14, 15, and 16. The training loss (Figure 14) decreases smoothly, indicating stable optimization. Validation macro-F1 (Figure 15) improves rapidly during early training and remains stable thereafter, suggesting effective convergence. The validation loss curve (Figure 16) exhibits moderate fluctuations, which are consistent with the increased variability introduced by augmented samples rather than overfitting.

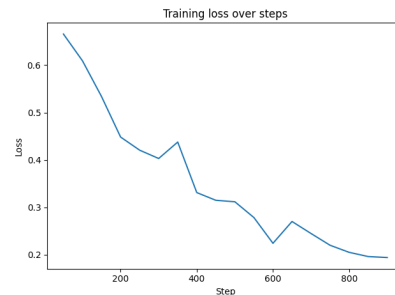


Figure 14: Training loss over steps for RoBERTa-large with data augmentation.

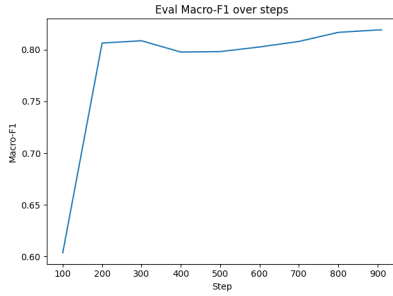


Figure 15: Validation macro-F1 over steps for RoBERTa-large with data augmentation.

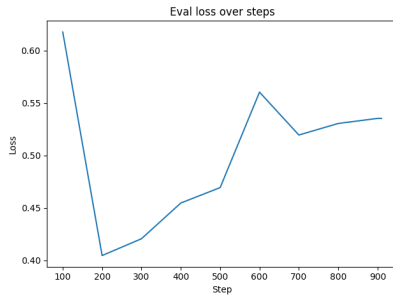


Figure 16: Validation loss over steps for RoBERTa-large with data augmentation.

GPT-4.1 Prompt-Based Classification. We additionally evaluate a prompt-based classification approach using GPT-4.1 in a zero-/few-shot setting on the English subset. This setup achieves an average accuracy of **0.61** and a macro-F1 score of **0.38**. Despite reasonable accuracy, the low macro-F1 indicates poor balance between classes, suggesting that the model struggles to reliably distinguish polarized from non-polarized content.

These results highlight the limitations of prompt-based large language models for this task when used without task-specific fine-tuning. In particular, predictions are sensitive to prompt formulation and tend to favor the majority class, leading to unstable and inconsistent performance compared to fine-tuned encoder models.

Sentence-Embedding Models. We evaluated a sentence-level approach based on the sentence-transformers/all-mpnet-base-v2 model using the 5-fold cross-validation protocol. Across the five folds, the model achieved macro-F1 scores of 0.803, 0.793, 0.792, 0.782, and 0.826, respectively. This resulted in a mean macro-F1 of 0.799 with a standard deviation of 0.015, indicating relatively stable performance across folds. While this sentence-embedding-based approach does not outperform the best token-level models in the

English setting, its consistent results suggest that it captures relevant global semantic information and serves as a strong baseline for further multilingual and hybrid modeling experiments.

Best Model. Based on the results of our exploratory fusion experiments, the hybrid architecture combining token-level representations from RoBERTa with sentence-level embeddings from MPNet emerged as our best-performing model. Using a 5-fold cross-validation protocol, the hybrid model achieved fold-level macro-F1 scores of 0.884, 0.904, 0.909, 0.920, and 0.878, resulting in a mean macro-F1 of 0.899 with a standard deviation of 0.016. This indicates both strong overall performance and stable behavior across folds.

When evaluated on the held-out test set, the hybrid model achieved a macro-F1 score of approximately 0.83, confirming that the gains observed during cross-validation generalize beyond the training data. In addition to macro-F1, the model attained an accuracy of approximately 0.88, with balanced precision and recall across classes. Specifically, precision values exceeded 0.88 for both classes, while recall remained consistently high, reaching over 0.91 for the non-polarized class and approximately 0.81–0.90 for the polarized class.

The confusion matrices further indicated that the hybrid model substantially reduces false negatives for the polarized class compared to sentence-level or token-level models alone, while maintaining strong performance on the majority class. These results suggest that integrating global sentence-level semantics from MPNet with fine-grained token-level contextual representations from RoBERTa enables a more robust decision boundary, particularly for detecting minority-class instances.

Result Comparison. Table 2 summarizes the performance of a wide range of models and configurations evaluated on the validation set. Traditional machine learning baselines, such as Naive Bayes with count features and logistic regression with TF-IDF, provide reasonable but clearly inferior performance compared to transformer-based approaches, with macro-F1 scores below 0.70. Prompt-based inference using GPT-4.1 performs substantially worse, indicating that zero-shot prompting is insufficient for this task without task-specific fine-tuning.

Among transformer models, we observe a consistent improvement when scaling model capacity and incorporating data augmentation. RoBERTa-based

models exhibit strong and stable performance, with RoBERTa-large outperforming its base variant and further gains obtained through augmentation strategies. In particular, RoBERTa-large with back-translation achieves the highest validation performance among non-cross-validation settings, reaching a macro-F1 of 0.844.

Cross-validation experiments further highlight the relative strengths of different architectures. While DeBERTa V3 models achieve competitive macro-F1 scores on the validation folds, with DeBERTa V3 large reaching 0.821, these improvements do not consistently translate to better generalization on the held-out test set. In fact, despite slightly stronger validation performance in some configurations, DeBERTa models were consistently outperformed by RoBERTa-large by over one percentage point on the test set across all comparable settings.

A similar discrepancy between validation and test performance is observed for augmentation-based models. Although RoBERTa-large with back-translation achieves a macro-F1 of 0.844 on the validation set, this configuration attains only 0.812 macro-F1 on the test set, indicating a degree of overfitting to the validation data. Notably, the *RoBERTa-large + LLM augmentation* configuration achieves a macro-F1 score of 0.935 under cross-validation on the validation data; however, this strong validation performance does not translate to comparable generalization on the test set, where the same model attains a macro-F1 of approximately 0.79. This further emphasizes the importance of evaluating generalization performance beyond validation metrics alone.

Overall, these results suggest that strong validation performance does not necessarily guarantee superior test performance. RoBERTa-large-based models demonstrate more robust generalization compared to alternatives, and the hybrid model achieves the highest macro-F1 under cross-validation. Consequently, our final model selection prioritizes architectures that exhibit consistent performance across validation, cross-validation, and test evaluations, rather than those that optimize validation metrics in isolation.

Additional Monolingual Setting Motivated by the strong performance of RoBERTa-large on English, we evaluated a comparable large-scale Chinese model, `hfl/chinese-roberta-wwm-ext-large`. The model obtained an F1 score of 91.64 on the test

set and 0.883 on the development set, suggesting that RoBERTa-style pretraining with whole-word masking is highly effective for Chinese as well.

Setup	Accuracy	Macro-F1
Naive Bayes (Count)	0.712	0.686
Logistic Regression (TF-IDF)	0.719	0.652
GPT-4.1 Prompt-Based	0.61	0.38
RoBERTa-base	0.79	0.76
RoBERTa-large	0.83	0.81
RoBERTa-large + LLM aug	0.85	0.85
RoBERTa-large + backtrans	0.88	0.844
MPNet + fold		0.79
DeBERTa V3 base + fold		0.804
DeBERTa V3 large + fold		0.821
RoBERTa-large + fold		0.8159
RoBERTa-large + LLM aug + fold		0.935
Hybrid Model + fold		0.89

Table 2: Results of Different Configurations on Validation Data

5.4 Multilingual Results

Table 3 presents per-language macro-F1 scores on the test set for the multilingual models under the different experimental configurations. Overall, the XLM-RoBERTa baseline demonstrates competitive performance across a wide range of languages, with particularly strong results for high-resource languages such as Persian, Nepali, Punjabi, and Chinese. However, performance remains substantially lower for several low-resource or morphologically complex languages, including Khmer and Italian, highlighting the inherent difficulty of multilingual polarization detection.

The refined XLM-RoBERTa configuration yields improvements for a subset of languages, most notably Arabic, English, Persian, Polish, and Turkish. At the same time, these gains are not uniform: for some languages, such as German and Italian, performance slightly decreases compared to the baseline. This variability suggests that global optimization strategies may benefit certain language groups while being less effective for others, likely due to differences in data size, linguistic structure, and label distribution.

Results obtained on the language subset largely follow the same trends observed in the full multilingual setting. For English, Italian, and Polish, performance remains comparable to that of the corresponding full multilingual configuration, indicating that the refined model retains reasonable robustness when trained on a smaller set of languages. Furthermore, performance improvements are particularly pronounced for Russian and Spanish, with Russian showing the most substantial gains among the subset languages. Nevertheless, the absence of consistent improvements across all subset languages

Experiment	AM	AR	BE	DE	EN	FA	HA	HI	IT	KH	MV	NE	OR	PA	PO	RU	SP	SW	TE	TU	UK	ZH
XLM-R - Baseline	0.7099	0.7361	0.8239	0.7184	0.7541	0.8374	0.7842	0.8171	0.6442	0.5505	0.3453	0.35	0.7594	0.8699	0.7715	0.7374	0.7088	0.7907	0.8643	0.7123	0.7532	0.8738
XLM-R - Modified Config	0.727	0.7822	0.84	0.6834	0.7824	0.8843	0.7842	0.7562	0.5931	0.5778	0.8098	0.8399	0.7037	0.8599	0.8092	0.7483	0.6776	0.7593	0.8559	0.7561	0.7415	0.8785
XLM-R - Modified Config (Subset)	—	—	—	0.6869	0.7721	—	—	—	0.6528	—	—	—	—	—	0.7918	0.8002	0.7272	—	—	—	—	—
MPNet - Baseline	0.734	0.7718	0.8373	0.6997	0.7884	0.8874	0.7405	0.8032	0.6193	0.5627	0.8598	0.89	0.7375	0.8397	0.7519	0.7991	0.6787	0.7987	0.8273	0.72	0.7388	0.883
MPNet - Modified Config	0.7425	0.7681	0.8329	0.6961	0.8122	0.8602	0.7443	0.8006	0.6458	0.5594	0.8516	0.8599	0.7485	0.8298	0.7146	0.7374	0.6463	0.8104	0.8557	0.7477	0.6856	0.8831
mDeBERTa-v3-base weighted training	0.762	0.8091	0.816	0.6971	0.8133	0.8274	0.7606	0.8293	0.6324	0.6846	0.8254	0.8292	0.7602	0.8091	0.8091	0.7214	0.7015	0.7707	0.805	0.8173	0.71	0.8551
mDeBERTa-v3-base ensemble Focal Loss	0.7347	0.8071	0.8336	0.6997	0.7803	0.8424	0.7534	0.8803	0.6418	0.673	0.8734	0.8296	0.7642	0.79	0.7918	0.7312	0.6776	0.7994	0.8559	0.8261	0.7224	0.9019
Ensemble + Augmentation	0.7158	0.7848	0.8277	0.7014	0.7922	0.8739	0.8034	0.8375	0.6793	0.6394	0.8725	0.8496	0.8086	0.7698	0.804	0.7481	0.6822	0.7904	0.8133	0.8172	0.7277	0.883

Table 3: Per-language performance comparison across different multilingual training configurations.

further underscores the challenges of balancing multilingual generalization with language-specific optimization.

The sentence-embedding-based MPNet models exhibit competitive performance across several languages, particularly English and Chinese, where they match or exceed the XLM-RoBERTa baseline. However, their performance is more uneven across languages, with notable drops for lower-resource settings. This behavior suggests that while sentence-level semantic representations can be effective in multilingual contexts, they may be more sensitive to data scarcity and language-specific characteristics than token-level multilingual encoders.

Taken together, the results in Table 3 indicate that no single multilingual configuration consistently outperforms the others across all languages. Instead, performance varies substantially depending on both the modeling approach and the target language, highlighting the importance of considering language-specific behavior when designing and evaluating multilingual polarization detection systems.

In addition to the XLM-RoBERTa and MPNet configurations, Table 3 also reports results for more advanced training strategies based on mDeBERTa-v3 and ensemble models. The mDeBERTa-v3-base model trained with class-weighted loss achieves competitive performance across several languages, particularly Amharic, Arabic, English, Hindi, Polish, and Turkish, indicating that explicit re-weighting can be beneficial for handling class imbalance in multilingual settings. However, its performance remains uneven across languages, with comparatively lower scores for languages such as Italian and German.

The ensemble variant of mDeBERTa-v3-base trained with focal loss further improves performance for a subset of languages, most notably Hindi, Chinese, Burmese, and Turkish, suggesting that focal loss can help emphasize harder or under-represented examples. Nevertheless, these gains are not consistent across all languages, and in some cases performance slightly degrades compared to

the weighted-loss variant. This variability indicates that while focal loss can be effective in multilingual contexts, its benefits are highly language-dependent and sensitive to data distribution.

Finally, the ensemble model combined with data augmentation yields strong results for several high-resource languages, including Persian, Hindi, Burmese, and Chinese, and achieves more balanced performance across a broader set of languages. At the same time, the augmented ensemble does not consistently outperform the best single-model configurations for all languages, and for some lower-resource languages the improvements remain limited. This observation aligns with earlier findings that augmentation-heavy and ensemble-based approaches may improve robustness for certain languages, but do not uniformly translate into superior generalization across the entire multilingual spectrum.

Overall, the results of the final configurations reinforce the conclusion that increasingly complex training strategies, such as loss re-weighting, focal loss, and ensembling, can yield localized improvements, but do not guarantee consistent gains across all languages. These findings highlight the trade-off between model complexity and cross-lingual robustness, and motivate the need for language-aware or adaptive multilingual modeling strategies.

6 Competition Ranking

Table 4 compares our per-language F1 scores with the best-performing submissions for each language. The system achieves top or near-top performance for several languages, including Persian, Hindi, Nepali, Punjabi, and Chinese, where the gap to the best system is minimal. In particular, for Persian, our model matches the best reported score.

For languages with lower leaderboard ranks, the absolute difference to the best-performing system often remains moderate, typically within a few F1 points. This indicates that leaderboard positions are frequently influenced by small score variations rather than substantial performance gaps. Overall, the results demonstrate robust and competitive multilingual performance across a diverse set of

languages.

Language	Our F1	Rank	Best F1	Δ
amh	0.7620	8	0.8057	0.0437
arb	0.8091	18	0.8801	0.0710
ben	0.8400	19	0.8760	0.0360
deu	0.7184	25	0.7922	0.0738
eng	0.8303	6	0.8902	0.0599
fas	0.8874	1	0.8874	0.0000
hau	0.8034	20	0.8578	0.0544
hin	0.8803	2	0.8882	0.0079
ita	0.6793	4	0.7036	0.0243
khm	0.6846	6	0.7069	0.0223
mya	0.8734	10	0.9147	0.0413
nep	0.8900	4	0.9000	0.0100
ori	0.8086	3	0.8531	0.0445
pan	0.8699	3	0.8796	0.0097
pol	0.7878	11	0.8620	0.0742
rus	0.8002	9	0.8308	0.0306
spa	0.7272	7	0.7515	0.0243
swa	0.8104	19	0.8509	0.0405
tel	0.8643	24	0.9406	0.0763
tur	0.8173	12	0.8783	0.0610
urd	0.7532	16	0.8094	0.0562
zho	0.9159	9	0.9439	0.0280

Table 4: Per-language F1 scores, leaderboard ranks, best reported scores, and absolute difference (Δ) between our system and the best-performing submission.

7 Conclusions

In this work, we investigated the problem of multilingual polarization detection in the context of the SemEval-2026 POLAR shared task. The dataset poses several challenges, including strong cross-lingual variation in label distributions, heterogeneous text lengths, and culturally grounded manifestations of polarization. Our exploratory analysis highlighted the limited predictive power of surface-level features and motivated the use of contextual representations and imbalance-aware evaluation.

We adopted a staged experimental strategy, beginning with a controlled monolingual setting in English before extending to multilingual modeling. In the monolingual experiments, transformer-based models substantially outperformed bag-of-words baselines, demonstrating the importance of contextual semantics for capturing polarized discourse. Scaling model capacity and incorporating data augmentation further improved performance, particularly for the minority class. Exploratory experiments with prompt-based large language mod-

els revealed that, without task-specific fine-tuning, such approaches remain unstable and underperform encoder-based models.

In the multilingual setting, we evaluated a range of transformer architectures and training strategies, including weighted loss functions, focal loss, data augmentation, and ensembling. Our results show that no single configuration consistently dominates across all languages: while some strategies yield improvements for specific languages, gains are highly language-dependent and sensitive to data distribution. This variability underscores the difficulty of designing universally optimal multilingual systems for polarization detection.

Overall, our findings suggest that robust polarization detection benefits from contextual modeling, careful handling of class imbalance, and evaluation with macro-averaged metrics. Future work will focus on reducing the strong language-specific performance variability observed in our multilingual experiments. A concrete next step is to introduce language-specific adapter layers into a shared multilingual encoder, where the base model is frozen and a separate lightweight adapter is trained for each language. This allows the model to specialize to language-specific expressions of polarization without retraining all parameters. Performance will be compared directly against joint multilingual fine-tuning under identical data splits and evaluation settings.

In addition, we plan to evaluate a language-routed mixture-of-experts architecture, where inputs are routed to language-specific classification heads based on known language identifiers rather than learned routing. This controlled setup enables direct analysis of whether explicit specialization reduces cross-language interference.

References

- Aymé Arango, Jorge Pérez, and Barbara Poblete. 2021. [Cross-lingual hate speech detection based on multilingual domain-specific word embeddings](#). *Preprint*, arXiv:2104.14728.
- Alexis Conneau, Kartikay Khandelwal, Naman Goyal, Vishrav Chaudhary, Guillaume Wenzek, Francisco Guzmán, Edouard Grave, Myle Ott, Luke Zettlemoyer, and Veselin Stoyanov. 2020. [Unsupervised cross-lingual representation learning at scale](#). *Preprint*, arXiv:1911.02116.
- Haixing Dai, Zhengliang Liu, Wenxiong Liao, Xiaoke Huang, Yihan Cao, Zihao Wu, Lin Zhao, Shaochen Xu, Wei Liu, Ninghao Liu, Sheng Li, Dajiang Zhu,

Hongmin Cai, Lichao Sun, Quanzheng Li, Dinggang Shen, Tianming Liu, and Xiang Li. 2023. [Aug-gpt: Leveraging chatgpt for text data augmentation](#). *Preprint*, arXiv:2302.13007.

Bosheng Ding, Linlin Liu, Lidong Bing, Canasai Kruengkrai, Thien Hai Nguyen, Shafiq Joty, Luo Si, and Chunyan Miao. 2020. [DAGA: Data augmentation with a generation approach for low-resource tagging tasks](#). In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 6045–6057, Online. Association for Computational Linguistics.

Long Kang, Jiaqi Yao, Ruoshuang Du, Lu Ren, Haifeng Liu, and Bo Xu. 2025. [A stance detection model based on sentiment analysis and toxic language detection](#). *Electronics*, 14(11).

Gary King and Langche Zeng. 2001. Logistic regression in rare events data. *Political Analysis*, 9(2):137–163.

Tsung-Yi Lin, Priya Goyal, Ross Girshick, Kaiming He, and Piotr Dollár. 2018. [Focal loss for dense object detection](#). *Preprint*, arXiv:1708.02002.

Usman Naseem, Juan Ren, Saba Anwar, Sarah Kohail, Rudy Alexandro Garrido Veliz, Robert Geislinger, Aisha Jabr, Idris Abdulmumin, Laiba Qureshi, Aarushi Ajay Borkar, Maryam Ibrahim Mukhtar, Abinew Ali Ayele, Ibrahim Said Ahmad, Adem Ali, Martin Semmann, Shamsuddeen Hassan Muhammad, and Seid Muhie Yimam. 2026. [Polar: A benchmark for multilingual, multicultural, and multi-event online polarization](#). *Preprint*, arXiv:2505.20624.

Raquel Rodriguez-Garcia and Roberto Centeno. 2024. [HAMiSoN-MTL at ClimateActivism 2024: Detection of hate speech, targets, and stance using multi-task learning](#). In *Proceedings of the 7th Workshop on Challenges and Applications of Automated Extraction of Socio-political Events from Text (CASE 2024)*, pages 89–95, St. Julians, Malta. Association for Computational Linguistics.